RESEARCH ARTICLE

Silymarin in non-cirrhotics with non-alcoholic steatohepatitis: A randomized, double-blind, placebo controlled trial

Victor J. Navarro[1]*, Steven H. Belle[2], Massimo D'Amato[3], Nezam Adfhal[4], Elizabeth M. Brunt[5], Michael W. Fried[6], K. Rajender Reddy[7], Abdus S. Wahed[2], Stephen Harrison[8], on behalf of the Silymarin in NASH and C Hepatitis (SyNCH) Study Group

1 Department of Digestive Disease and Transplantation, Einstein Medical Center and Sidney Kimmel Medical College, Philadelphia, Pennsylvania, United States of America, **2** Department of Epidemiology, University of Pittsburgh Graduate School of Public Health, Pittsburgh, Pennsylvania, United States of America, **3** Clinical Research, Rottapharm Biotech, Monza MB, Italy, **4** Division of Hepatology, Department of Medicine, Beth-Israel Deaconess Medical Center, Boston, Massachusetts, United States of America, **5** Department of Pathology and Immunology, Washington University School of Medicine, CB 8118, St. Louis, Missouri, United States of America, **6** Department of Medicine, Division of Gastroenterology and Hepatology, Liver Center, University of North Carolina, Chapel Hill, North Carolina, United States of America, **7** Department of Medicine, Division of Gastroenterology University of Pennsylvania, Philadelphia, Pennsylvania, United States of America, **8** Department of Medicine, Division of Gastroenterology, Brooke Army Medical Center, Fort Sam Houston, Texas, United States of America

These authors contributed equally to this work.
* navarrov@einstein.edu

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Abstract

The botanical product silymarin, an extract of milk thistle, is commonly used by patients to treat chronic liver disease and may be a treatment for NASH due to its antioxidant properties. We aimed to assess the safety and efficacy of higher than customary doses of silymarin in non-cirrhotic patients with NASH. This exploratory randomized double-blind placebo controlled multicenter Phase II trial tested a proprietary standardized silymarin preparation (Legalon®, Rottapharm|Madaus, Mylan) and was conducted at 5 medical centers in the United States. Eligible adult patients had liver biopsy within 12 months showing NASH without cirrhosis with NAFLD Activity Score (NAS) $\geq$4 per site pathologist's assessment. Participants were randomized to Legalon® 420 mg, 700 mg, or placebo t.i.d. for 48 weeks. The primary endpoint was histological improvement $\geq$2 points in NAS. Of 116 patients screened, 78 were randomized. There were no significant differences in adverse events among the treatment groups. After 48–50 weeks, 4/27 (15%) in the 700 mg dose, 5/26 (19%) participants randomized to 420 mg, and 3/25 (12%) of placebo recipients reached the primary endpoint ($p = 0.79$) among all randomized participants, indicating no benefit from silymarin in the intention to treat analysis Review by a central pathologist demonstrated that a substantial number of participants (49, 63%) did not meet histological entry criteria and that fibrosis stage improved most in the placebo treated group, although not significantly different from other groups. Silymarin (Legalon®) at the higher than customary doses tested in this study is safe and well tolerated. The effect of silymarin in patients with NASH remains inconclusive due to the substantial number of patients who entered the study but did not meet entry histological criteria, the lack of a statistically significant improvement in NAS of silymarin treated patients, and the unanticipated effect of placebo on fibrosis indicate the need for additional clinical trials.

Trial Registration: clinicaltrials.gov, Identifier: NCT00680407.

Nezam Afdhal- 1U01AT003571; Michael W. Fried- 1U01AT003560; K. Rajender Reddy- 1U01AT003573; sponsorship was assumed by Rottapharm|Madaus (now Mylan) during the course of the trial. Stephen Harrison joined after transfer of the study to Rottapharm/Madaus. In addition, Rottapharm|Madaus provided silymarin and placebo. No conflicts of interest exist as this was an investigator-initiated trial. Mylan, formerly Rottapharm|Madaus, manufacturer of the silymarin formulation used in this study, as corporate entity had no direct or indirect role nor involvement in the design of the trial, data collection or preparation or submission of the manuscript for publication of this registered (http://clinicaltrials.gov/ct2/show/NCT00680407) investigator-initiated trial. None of the authors has a personal conflict of interest with the manufacturer of any of the marketed silymarin formulations. Dr. Massimo D'Amato was an employee of the Italian subsidiary of Rottapharm|Madaus at the time of the study, but left Rottapharm|Madaus to join Rottapharm Biotech in 2014 and participated in the preparation of the manuscript as an independent scientist. Rottapharm Biotech is an independent company from Mylan, does not market Legalon nor any sylimarin formulation and as corporate entity had no role in the conception, design and performance of this study, nor in the decision to submit the manuscript for publication.

Competing interests: The authors confirm the following competing interests: Victor Navarro, M.D. No conflicts relevant to this trial; Steven H. Belle, Ph.D. No conflicts relevant to this trial; Massimo D'Amato, M.D. Dr. D'Amato was an employee of the Italian subsidiary of the Rottapharm|Madaus, now Mylan, manufacturer of the silymarin formulation used in the study, at the time of the study, but left Rottapharm|Madaus to join Rottapharm Biotech in 2014 and participated in the preparation of the manuscript as an independent scientist. Rottapharm Biotech is an independent company from Mylan, does not market Legalon nor any sylimarin formulation and as corporate entity had no role in the conception, design and performance of this study, nor in the decision to submit the manuscript for publication; Nezam Afdhal, M.D. Dr. Afdhal serves as a consultant/advisory board member for Merck, Gilead, Echosens, Ligand, Jannsen, and Shionogi. He is Director, TRIO Heathcare. He owns stock or stock options in Sprinbank, Allurion, and TRIO; Elizabeth M. Brunt, M.D. Dr. Brunt was compensated by Rottapharm|Madaus for her review of the liver biopsies; Michael W. Fried, M.D. Dr. Fried receives research grants paid to his institution from AbbVie, BMS, Gilead, Merck. He serves as an unpaid

Introduction

Non-alcoholic fatty liver disease (NAFLD) and its progressive form, non-alcoholic steatohepatitis (NASH) characterized by steatosis and necroinflammation, with or without centrilobular fibrosis, have emerged as prevalent problems in Western populations. The main risk factors for developing NAFLD and NASH are components of the metabolic syndrome; increased weight, insulin resistance, hypertension, and hyperlipidemia. Liver biopsy remains the gold standard for diagnosing and assessing the degree of injury in NASH [1,2]. Although exercise and weight loss, surgical bariatric procedures, and pharmacological interventions, including insulin sensitizing agents and Vitamin E, have shown promise in treating people with NAFLD or NASH, there is no approved therapy for these disorders. There is ongoing public health concern since it is estimated that up to 30% of the US population has NAFLD, and another one third has NASH.

Silymarin, an extract of milk thistle (*Silybum marianum*), is the botanical treatment most commonly used for liver disorders in the United States, owing to its purported hepatoprotective properties [3]. Through its antioxidant properties, silymarin may mitigate lipid peroxidation and the production of free radical injury, a suspected mechanism of liver injury in NASH. Studies evaluating the use of silymarin in this capacity have found it to be effective in scavenging hydroxyl radicals, preventing the release of TNF alfa, and restoring normal levels of superoxide dismutase, a precursor of glutathione [4,5,6,7].

Recently, Kheong and colleagues reported that Silymarin at a single dose was safe and well tolerated in a Malayasian NASH population, but did not result in a statistically significant reduction in the NAS compared with placebo.[8] In view of the limited data available on dosing and pharmacokinetics of silymarin, an initial dose-ranging study [9] was performed to identify adequate silymarin doses to be tested in proof of concept studies, including the current trial. Here, we aimed to confirm the safety and preliminarily assess the efficacy of silymarin in patients with biopsy confirmed NASH without cirrhosis.

Materials and methods

Trial design

The "Silymarin in NASH and C Hepatitis (SyNCH)" study was a randomized, double-blind, placebo controlled phase II multicenter trial to evaluate the safety and explore the efficacy of 2 doses of a standardized form of silymarin (Legalon®, Rottapharm|Madaus, Mylan) compared with placebo. Although the SyNCH study comprised two patient poulations, those with Hepatitis C [10] and NASH, the current study pertains only to the latter population. Diabetic and non-diabetic patients with NASH and without cirrhosis were randomized to 3 treatment groups for up to 48-50-weeks treatment duration. Enrollment began in May 2008, and was completed in August 2011, with follow-up completed in November 2012. The trial was initially funded as a cooperative agreement award between the National Center for Complementary and Alternative Medicine and the National Institutes for Diabetes, Digestive and Kidney

consultant to AbbVie, BMS, Merck, and TARGET PharmaSolutions and owns stock in TARGET PharmaSolutions, which is held in an independently managed trust; K. Rajender Reddy, M.D. Dr. Reddy serves on the Scientific Advisory Board-Gilead, Merck, Abbvie, BMS, Spark Therapeutics. Research Support (Paid to the University of Pennsylvania)-Gilead, Merck, Abbvie, BMS, Conatus, Mallinckrodt, Intercept, Exact Sciences. DSMB-Novartis; Abdus S. Wahed, PhD: No conflicts relevant to this trial; Stephen Harrison, M.D. Consultant/Advisory board for Prometic, Innovate, Gelesis, CiVi, Contravir, Cymabay, Galectin, Northsea, Hightide, Madrigal, Metacrine, NGM, Cirius, Akero, Terns, Viking, Blade, Poxel, 3v Bio, Axcella, Merck, Gilead, Intercept, Genfit, Novo Nordisk, Echosens, Perspectum, Novartis and Medpace. Research support from Gilead, Novo Nordisk, BMS, Pfizer, Novartis, Cymabay, NGM, Axcella, Hightide, Conatus, Galectin, Intercept, Genfit, Akero, Metacrine, Cirius, Contravir, Madrigal, Enyo, Northsea. This does not alter our adherence to PLOS ONE policies on sharing data and materials.

Abbreviations: ALT, Alanine Aminotransferase; ANOVA, Analysis of Variance; AST, Aspartate Aminotransferase; DSMB, Data and Safety Monitoring Board; HOMAr, Homeostatic Model Assessment; LOCF, Last Observation Carried Forward; NAFLD, Non-Alcoholic Fatty Liver Disease; NAS, Non-Alcoholic Fatty Liver Disease Activity Score; NASH, Non-Alcoholic Steatohepatitis.

Diseases. However, in May 2009, funding of the study was transferred to the sponsor, Rottapharm|Madaus, as an investigator-initiated clinical trial.

Participants

Patients over 18 years of age with AST or ALT $>$ 40 IU/L within one year of screening and at least once during a 30-day screening period, and with suspected NAFLD were eligible for the study. At screening, patients were counseled to follow a healthy diet and lifestyle. Dietary recommendations included a decrease in saturated fats as well as total fats to $<$30% of total calories and macronutrient distribution of 45 to 55% carbohydrate, 25 to 35% fat and 15 to 20% protein. Patients were provided with dietary counseling to maintain glycemic control as well as to maintain a target weight/BMI that reflects no more than a +/- 10% change of body weight. Liver biopsy within 12 months of randomization confirming NASH was required for entry; the absence of cirrhosis and a NAFLD Activity Score (NAS) of 4 or greater on the baseline biopsy as read by a site pathologist qualified patients for the study. Patients meeting study entry criteria were stratified by the presence or absence of diabetes.

Patients were excluded if they had evidence for other chronic liver diseases or decompensation, a history of immunologically mediated liver disease or other severe medical illnesses, if they refused to adhere to limitations on alcohol consumption (average alcohol consumption of more than 1 drink per day or more than 2 drinks on any one day over the 30 days prior to the screening period), were diabetic with any change in anti-diabetic medication during the screening period, or had poor control of their diabetes as indicated by HbA1c $>$ 8%. Secretagogues (sulfonylureas) and metformin were not permitted, given their purported impact on NAFLD, and anti-hyperlipidemics were permitted. In addition, patients with BMI $>$ 45 kg/m^2 were excluded, and weight must have been shown to be stable, with no more than a 10% change between the baseline liver biopsy and enrollment. Patients were ineligible if they had used other milk thistle preparations for a period of 90 consecutive days or longer between biopsy and initial screening, or within 30 days prior to screening if the liver biopsy was performed during the screening period. Patients were also excluded if they had used other antioxidants such as vitamin E, vitamin C, glutathione, alpha-tocopherol, or non-prescribed complementary alternative medications (including dietary supplements, megadose vitamins, herbal preparations, and special teas) within 30 days prior to screening. Medications known to produce a NAFLD or NASH-like histological picture (eg; methotrexate) were not permitted.

Participants were recruited at 5 clinical sites in the United States. The study was approved by the institutional review boards at Thomas Jefferson University, Beth Israel Deaconness Medical Center, University of North Carolina-Chapel Hill, University of Pennsylvania, the Brooke Army Medical Center, and at the Data Coordinating Center (DCC, University of Pittsburgh). All patients provided written informed consent. An independent Data and Safey Monitoring Board (DSMB) approved the initial protocol and regularly reviewed the study progress.

Interventions and outcomes assessment

Participants were randomly assigned by the Data Coordinating Center at the University of PIttsburgh to 1 of 3 treatment groups: Legalon® 420 mg or 700 mg, or placebo administered three times daily for up to 48–50 weeks. Legalon® is a proprietary milk thistle seed extract standardized to a silymarin content of 140 mg per capsule. Treatment could be extended for up to 54 weeks following randomization to ensure that the post-treatment liver biopsy was performed while the participant was still taking study drug. After completing therapy, participants were monitored for an additional 12 weeks.